AI/ML EXPERIMENT REPORT

Phase 2: L-NER Pipeline & Distillation
Phase 3: Rhetorical Role Prediction (RR)

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Scope: Reconstruction from the model-related details currently available in conversation context.

Evidence note
This report distinguishes recorded configuration and metric values from planned work and unavailable details. It is not an original training log, and it does not claim that unverified training runs completed.

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Executive Summary

This report consolidates the available records for two separate competition phases. Phase 2 concerns a Legal Named Entity Recognition (L-NER) pipeline built around Architecture 4 (Arch4), including a planned teacher–student distillation workflow. Phase 3 concerns sentence-level Rhetorical Role Prediction (RR), implemented with an RRClassifier and LoRA.

Area	Recorded state	Key evidence
Phase 2 — L-NER	Pipeline described; distillation plan recorded	Arch4 configuration: 201,443,328 parameters; step 10,000; best validation loss 2.3814771915475528
Phase 3 — RR	Scoring completed successfully in a recorded run	Macro-F1 0.300542; macro-precision 0.309071; macro-recall 0.315324

Interpretation: Phase 3 has a concrete recorded evaluation result. Phase 2 has detailed pipeline and distillation design notes, but the surviving context does not include a definitive final Phase 2 leaderboard score or proof that the student distillation run completed.

Evidence and limitations

Values in this document come from the surviving conversation context. Original notebook files, raw logs, complete configurations, checkpoints, and deleted conversations were not available for independent verification during report generation. Unknown values are explicitly left unknown rather than inferred.

1. Phase 2 — L-NER Pipeline

1.1 Objective and workflow

The recorded Phase 2 workflow was described as a single-file pipeline that runs training, strict-F1 validation, prediction, and submission packaging. The script name retained in context is `lner_pipeline.py`, associated with the Arch4 checkpoint.

Stage	Recorded purpose / status
Train	Train the L-NER model using the selected architecture/checkpoint setup; exact command and configuration are unavailable.
Validate	Validation was described as using strict F1; the exact implementation and validation result are not retained.
Predict	Generate predictions for the evaluation/test inputs; prediction output filenames are unavailable.
Package	Create `submission.zip`; the surviving context does not confirm the final archive contents or score.

1.2 Architecture and checkpoint

Field	Recorded value
Architecture	Arch4
Exact recorded parameter count	201,443,328 (approximately 201.443M)
Training step	10,000
Best validation loss	2.3814771915475528
Checkpoint naming pattern	`arch4_lora_f{k}_best.pt` (fold-specific)
Loading status	Strict checkpoint loading was reported as successful
Attention pattern	`L L G L L G L L G L L G`

A separate distillation note described the teacher as approximately 205M parameters. The exact Arch4 configuration record is 201,443,328 parameters; the two figures are not identical and should not be silently conflated.

1.3 Teacher–student distillation plan

The recorded plan was to compress Arch4 from 12 transformer layers to a 6-layer student. The selected layer indices were `[0, 1, 2, 6, 7, 8]`. The stated motivation was to preserve the periodic global-attention structure in the reduced model.

The fold-specific setup stated that teacher for fold k and student for fold k were trained using the same 56 training documents. This is a recorded design note; the exact fold construction and validation boundaries are not available for audit.

The loss was described as a weighted combination of hard-label cross-entropy and temperature-scaled teacher-to-student KL divergence:

$$L = \alpha \times CE(\text{hard labels}) + (1 - \alpha) \times T^2 \times KL(\text{teacher} \blacksquare \text{student})$$

The values of α and T, optimizer, learning rate, batch size, training duration, and student evaluation metrics are not preserved. Therefore, the student result should be treated as unverified / unavailable.

1.4 Dataset and sequence-length context

Item	Recorded detail
Phase 1 dataset reference carried into model context	`phase1-tokens-v40k-clean1`
Vocabulary	40,000 tokens
Competition parameter constraint	50M–300M parameters (specification Section 7.5)
Compression/distillation reference	Specification Section 7.4
Official test IDs	1,000
Sequence-length constraint discussed	SRC_LEN + TGT_LEN = 1,024
Prompt lengths observed	Average approximately 609 tokens; examples up to approximately 918 tokens

The record says CFG['SRC_LEN'] and CFG['TGT_LEN'] were initially None and needed to be assigned valid values summing to 1,024. Their final individual values are unknown. The Phase 1 dataset/specification details are included because they were part of the model context associated with the L-NER pipeline, but the available notes do not fully establish which details apply to every Phase 2 run.

1.5 Debugging and implementation notes

• A dataset mount error was encountered. A later update said the checkpoint path had been corrected and the startup check succeeded.
• Prediction-part merging failed with an assertion error because `PART_FILES` was empty.
• Arch1 loader status was described as unverified; using it was considered risky. The retained note mentions a score around 0.17, but its precise metric and evaluation conditions are unknown.
• No complete stack trace, exact notebook URL, final submission ZIP, or definitive Phase 2 competition score is retained.

2. Phase 3 — Rhetorical Role Prediction (RR)

2.1 Task and dataset

Phase 3 is recorded as a sentence-level legal NLP task: Rhetorical Role Prediction. The dataset was identified as `Exploration-Lab/CS-781-Capstone`, with training data in `CL_train.json`.

Dataset field	Recorded value
Training documents	40
Total labeled sentences	10,593
Total sentence samples	16,845
CL test documents	5
IT test documents	5
CL test sentences	1,339
IT test sentences	858
Vocabulary size	40,000
PAD / BOS / EOS IDs	0 / 1 / 2

The official label vocabulary was mentioned in prior context, but the complete label names are not retained. The record does not explain the difference between the 10,593 labeled sentences and 16,845 sentence samples, so these counts are reported as given and should not be interpreted as interchangeable.

2.2 Model and parameter efficiency

Field	Recorded value
Model class	`RRClassifier`
Device	`cuda:0`
LoRA adapter	Present
LoRA target count	16
Trainable parameters	206,605
Total parameters	88,417,549
Trainable percentage	Approximately 0.234%

The trainable parameter count and percentage indicate that only a small fraction of the model's parameters were trainable in the recorded LoRA setup. The underlying backbone architecture, LoRA rank, alpha, dropout, target module names, and base checkpoint path are not available.

2.3 Evaluation results

Metric	Recorded value	Notes
Macro-F1	0.300542	Primary macro-averaged F1 result
Macro-precision	0.309071	Macro-averaged precision
Macro-recall	0.315324	Macro-averaged recall

The recorded scoring run completed successfully. The scores were written to `/app/output/scores.json`.

2.4 Evaluation information not retained

The following items are not available in the surviving context: per-class precision/recall/F1, confusion matrix, exact evaluation split used for the reported metrics, number of epochs, loss curve, early-stopping criteria, random seed, final leaderboard rank, and original prediction file.

3. Cross-phase comparison and interpretation

Dimension	Phase 2 — L-NER	Phase 3 — RR
Task type	Legal NER pipeline with planned compression	Sentence-level rhetorical-role classification
Model record	Arch4, 201,443,328 parameters	RRClassifier, 88,417,549 total parameters
Training/evaluation evidence	Step 10,000 and best validation loss recorded; final phase score unknown	Scoring completed; macro metrics recorded
Efficiency strategy	Planned 12-to-6-layer teacher–student distillation	LoRA; 206,605 trainable parameters
Main unresolved issue	Student training completion and final submission/score	Per-class diagnostics and full training configuration

> Do not compare Phase 2's validation loss directly with Phase 3's macro-F1. They measure different objectives and tasks. Likewise, Arch4's validation loss cannot establish the quality of the Phase 2 submission without the precise evaluation protocol and test results.

4. Recovery checklist and next actions

Recommended actions to recover the original evidence:

1. Open Kaggle Code and locate the notebooks associated with `lner_pipeline.py`, Arch4, distillation, and RR.
2. Inspect saved notebook versions, run logs, and output artifacts. Download any retained checkpoints, predictions, score JSON files, and training logs.
3. Search notebook outputs for `best validation loss`, `strict F1`, `scores.json`, `PART_FILES`, and checkpoint filenames.
4. For RR, recover `/app/output/scores.json` or the notebook cell that generated it, then collect per-class reports if available.
5. Rebuild a run registry with one row per run: run ID, dataset version, split, architecture, parameter count, checkpoint, seed, step/epoch, validation metrics, test metrics, and artifact path.
6. Mark every missing field as unknown until it is verified from an original artifact.

5. Known gaps and confidence

Information	Current status
Arch4 parameter count, step, best loss	Recorded in surviving context; original logs unavailable for verification
Phase 2 pipeline name and intended stages	Recorded; completion of final submission is not confirmed
Distillation layer selection and loss form	Plan recorded; α, T and final student metrics unknown
RR model parameter counts and macro metrics	Recorded values; original scores file not available here
Original deleted ChatGPT messages	Not recoverable from the current context
Kaggle notebooks/checkpoints/logs	Not inspected; recovery depends on whether versions and artifacts still exist

Conclusion

The strongest numerical evidence available for Phase 2 is the Arch4 configuration (201,443,328 parameters), training step 10,000, and best recorded validation loss of 2.3814771915475528. The teacher–student compression strategy is documented as a plan, but its final outcome is unknown. For Phase 3, the RRClassifier configuration and completed scoring metrics are recorded: Macro-F1 0.300542, macro-precision 0.309071, and macro-recall 0.315324. The next priority is to retrieve Kaggle notebook versions and output artifacts to validate and extend this reconstructed report.

Report status: reconstructed from available conversation context; not independently verified against original logs or files.